

MCTR II

Spooling

Work Package Scope of Work — Engineering Design 2026

CONSTRAINT: No worm gearbox

Main Work Elements

1. **Hoist gearbox spec and dev** — minimum 2-stage gearbox for the spool, driven off EE1's induction motor, no worm gearbox permitted.
2. **Active brake design** — fail-safe holding brake satisfying the inherently-safe requirement, its control interface back to the controller, and selection of its own COTS low-voltage supply rail.
3. **On-hook load measurement** — load cell in the hoist path, feeding telemetry and interlock use (via an ADS1220 differential channel or a dedicated HX711 on the spare IOs — not yet committed).
4. **Dynamics simulation** — lift/lower dynamics — load, speed, gear ratio, and brake engagement behaviour — validating the mechanism against required performance.
5. **Controller and safety interface** — wiring load cell and brake signals into the controller, and defining interlock logic tying brake engagement to load and fault conditions.

Scoped for effort parity with the other four 2026 work packages (MCTR I, MCTR II, EEI I, EEI II, REI I).